VE216 Mid-2

Dirichlet Conditions

For FS ① bounded

② finite max/min in one period

③ finite discontinuities in one period

For FT ① absolutely integrable $\int_{-\infty}^{\infty}|x(t)|dt<\infty$

② finite max/min/discontinuities in any finite interval

③ size of discontinuities is finite

* $\hat{x}(t)$ converges to the midpoint if dis~

Trigonometric FS

$x(t)=B[0]+\sum_{k=1}^{\infty}B[k]\cos(k\omega_0 t)+A[k]\sin(k\omega_0 t)$

$$\begin{cases} B[0]=\frac{1}{T}\int_0^T x(t)dt \\ B[k]=\frac{2}{T}\int_0^T x(t)\cos(k\omega_0 t)dt \\ A[k]=\frac{2}{T}\int_0^T x(t)\sin(k\omega_0 t)dt \end{cases} \qquad \begin{cases} B[k]=X[k]+X[-k] \\ A[k]=j(X[k]-X[-k]) \end{cases}$$

Properties of FT (略)

Time reversal $x(-t)\xleftrightarrow{FT}X(-j\omega)$

Conjugation $x^*(t)\xleftrightarrow{FT}X^*(-j\omega)$

Even-odd decomp $\begin{cases} x_e(t)\xleftrightarrow{FT}Re\{X(j\omega)\} \\ x_o(t)\xleftrightarrow{FT}Im\{X(j\omega)\} \end{cases}$

How to 求 System Frequency Response

[I] $y(t)=h(t)*x(t)\xleftrightarrow{FT}Y(j\omega)=H(j\omega)X(j\omega)$

[II] $\sum_{k=0}^{N}a_k\frac{d^k}{dt^k}y(t)=\sum_{k=0}^{M}b_k\frac{d^k}{dt^k}x(t)$

$\xleftrightarrow{FT}\sum_{k=0}^{N}a_k(j\omega)^kY(j\omega)=\sum_{k=0}^{M}b_k(j\omega)^kX(j\omega)$

$$\Rightarrow H(j\omega)=\frac{Y(j\omega)}{X(j\omega)}=\frac{\sum_{k=0}^{M}b_k(j\omega)^k}{\sum_{k=0}^{N}a_k(j\omega)^k}=\frac{c_1}{j\omega-d_1}+\cdots+\frac{c_N}{j\omega-d_N}$$

$$=\sum_{k=0}^{N}e^{d_k t}(d_k<0)$$

special cases $\begin{cases} ① \deg Y>\deg X. \\ ② \text{complex roots} \\ ③ \text{multiple roots} \end{cases}$

法一: e.g. $X(j\omega)=\frac{A_1}{(j\omega+1)^2}+\frac{A_2}{j\omega+1}$

$A_1=(j\omega+1)^2X(j\omega)|_{j\omega=-1}$

$A_2=\frac{d}{d(j\omega)}\{(j\omega+1)^2X(j\omega)\}|_{j\omega=-1}$

法二: 令 $j\omega=0$ 求解.

Time-bandwidth Property

Duration $T_d=\sqrt{\frac{\int_{-\infty}^{\infty}t^2|x(t)|^2dt}{\int_{-\infty}^{\infty}|x(t)|^2dt}}$ time-domain; for slower response

Bandwidth $B_\omega=\sqrt{\frac{\int_{-\infty}^{\infty}\omega^2|X(j\omega)|^2d\omega}{\int_{-\infty}^{\infty}|X(j\omega)|^2d\omega}}$ frequency-domain; for wider bandpass

Uncertainty principle $T_dB_\omega\geq\frac{1}{2}$ or $\sigma_t^2\sigma_\omega^2\geq\frac{1}{4}$

$\int_{-\infty}^{\infty}e^{-x^2/2\sigma^2}dx=\sigma\sqrt{2\pi}.\ \sigma>0$ $\int_{-\infty}^{\infty}x^2e^{-x^2/2\sigma^2}dx=\sigma^3\sqrt{2\pi}.\ \sigma>0$	$\mathrm{sinc}\,x=\frac{\sin(\pi x)}{\pi x}$

Sampling

$x_s(t)=\sum_{n=-\infty}^{\infty}x[n]\delta(t-nT_s)=\sum_{n=-\infty}^{\infty}x(nT_s)\delta(t-nT_s)$

$=x(t)p(t)$ (impulse train)

FT: $X_s(j\omega)=\frac{1}{T_s}\sum_{k=-\infty}^{\infty}X(j(\omega-k\omega_s))$

$x[n]\xleftrightarrow{DTFS}X(e^{j\omega})=X_s(j\omega)|_{\omega=\frac{\Omega}{T_s}}$

Nyquist rate

Shannon Sampling Theorem: $\omega_s>2W\Rightarrow$ no aliasing

Ideal Reconstruction

$X(j\omega)=X_s(j\omega)H_r(j\omega)$ where $H_r(j\omega)=\begin{cases} T_s, & |\omega|\leq\omega_s/2 \\ 0, & |\omega|>\omega_s/2 \end{cases}$

$x(t)=x_s(t)*h_r(t)$, where $h_r(t)=\frac{T_s\sin(\frac{\omega_s}{2}t)}{\pi t}$

$=\sum_{n=-\infty}^{\infty}x[n]\sin(\omega_s(t-nT_s)/2\pi)$

Partial Reconstruction (Zero-order Holder)

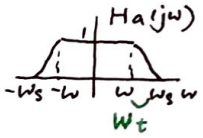
$h_0(t)=\begin{cases} 1, & 0<t<T_s \\ 0, & t<0, t>T_s \end{cases}\xleftrightarrow{FT}H_0(j\omega)=2e^{-j\omega T_s/2}\frac{\sin(\omega T_s/2)}{\omega}$

$H_c(j\omega)=\begin{cases} \frac{\omega T_s}{2\sin(\omega T_s/2)}, & |\omega|<\omega_m \\ 0, & |\omega|>\omega_s-\omega_m \end{cases}$

Oversampling

$\omega_s-W_s>W$

$(W_s=W_t+W)$ $W_t<\omega_s-2W$



Discrete-time Processing

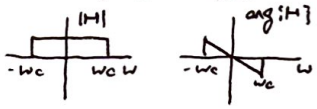
$x(t)\to\boxed{H_a(j\omega)}\xrightarrow{x_a(t)}\boxed{\text{Sample}}\xrightarrow{x[n]}\boxed{H}\xrightarrow{y[n]}\boxed{H_0(j\omega)}\xrightarrow{y_0(t)}\boxed{H_c(j\omega)}\to y(t)$

$G(j\omega)=[\frac{1}{T_s}H_0(j\omega)H_c(j\omega)H_a(j\omega)]H(e^{j\omega T_s})$

Ideal Low-pass Filter

$H(j\omega)=\begin{cases} e^{-j\omega t_0} & |\omega|\leq\omega_c \\ 0 & |\omega|>\omega_c \end{cases}\Rightarrow h(t)=\frac{\sin(\omega_c(t-t_0))}{\pi(t-t_0)}$

$=\frac{\omega_c}{\pi}\mathrm{sinc}(\frac{\omega_c}{\pi}(t-t_0))$



Proof of FT

$\int_{-\infty}^{\infty}x(t)e^{-j\omega t}dt=\int_{-\infty}^{\infty}[\frac{1}{2\pi}\int_{-\infty}^{\infty}X(jr)e^{jrt}dr]\cdot e^{-j\omega t}dt$

$=\int_{-\infty}^{\infty}[\frac{1}{2\pi}X(jr)\int_{-\infty}^{\infty}e^{j(r-\omega)t}dt]dr$

$=\int_{-\infty}^{\infty}\frac{1}{2\pi}X(jr)2\pi\delta(r-\omega)dr=X(j\omega)$

$\int_{-\infty}^{\infty}e^{-j\omega t}dt=\int_{-\infty}^{0}e^{-j\omega t}dt+\int_{0}^{\infty}e^{-j\omega t}dt$

$=\lim_{a\to0^+}(\int_{-\infty}^{0}e^{(a-j\omega t)}dt+\int_{0}^{\infty}e^{(-a-j\omega t)}dt)$

$=\lim_{a\to0^+}(\frac{1}{a-j\omega}+\frac{1}{a+j\omega})=\lim_{a\to0^+}\frac{2a}{a^2+\omega^2}=\begin{cases} \infty & \omega=0 \\ 0 & \omega\neq0 \end{cases}$

$\int_{-\infty}^{\infty}\frac{2a}{a^2+\omega^2}d\omega=\cdots=2\pi$

$\Rightarrow\int_{-\infty}^{\infty}e^{-j\omega t}dt=2\pi\delta(\omega)$

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Figure 1: FS

Time Domain	*Frequency Domain*
$x(t) = \sum_{k=-\infty}^{\infty} X[k]e^{jk\omega_o t}$ *Period* $= T$	$X[k] = \frac{1}{T}\int_0^T x(t)e^{-jk\omega_o t}\,dt$ $\omega_o = \frac{2\pi}{T}$
$x(t) = \begin{cases} 1, & \lvert t\rvert \le T_o \\ 0, & T_o < \lvert t\rvert \le T/2 \end{cases}$	$X[k] = \frac{\sin(k\omega_o T_o)}{k\pi}$
$x(t) = e^{jp\omega_o t}$	$X[k] = \delta[k-p]$
$x(t) = \cos(p\omega_o t)$	$X[k] = \frac{1}{2}\delta[k-p] + \frac{1}{2}\delta[k+p]$
$x(t) = \sin(p\omega_o t)$	$X[k] = \frac{1}{2j}\delta[k-p] - \frac{1}{2j}\delta[k+p]$
$x(t) = \sum_{p=-\infty}^{\infty}\delta(t-pT)$	$X[k] = \frac{1}{T}$

Figure 2: FT

Time Domain	*Frequency Domain*
$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{j\omega t}\,d\omega$	$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\,dt$
$x(t) = \begin{cases} 1, & \lvert t\rvert \le T_o \\ 0, & \text{otherwise} \end{cases}$	$X(j\omega) = \frac{2\sin(\omega T_o)}{\omega}$
$x(t) = \frac{1}{\pi t}\sin(Wt)$	$X(j\omega) = \begin{cases} 1, & \lvert\omega\rvert \le W \\ 0, & \text{otherwise} \end{cases}$
$x(t) = \delta(t)$	$X(j\omega) = 1$
$x(t) = 1$	$X(j\omega) = 2\pi\delta(\omega)$
$x(t) = u(t)$	$X(j\omega) = \frac{1}{j\omega} + \pi\delta(\omega)$
$x(t) = e^{-at}u(t), \quad \mathrm{Re}\{a\} > 0$	$X(j\omega) = \frac{1}{a+j\omega}$
$x(t) = te^{-at}u(t), \quad \mathrm{Re}\{a\} > 0$	$X(j\omega) = \frac{1}{(a+j\omega)^2}$
$x(t) = e^{-a\lvert t\rvert}, \quad a > 0$	$X(j\omega) = \frac{2a}{a^2+\omega^2}$
$x(t) = \frac{1}{\sqrt{2\pi}}e^{-t^2/2}$	$X(j\omega) = e^{-\omega^2/2}$

Figure 3: FT of Periodic

Periodic Time-Domain Signal	*Fourier Transform*
$x(t) = \sum_{k=-\infty}^{\infty} X[k]e^{jk\omega_o t}$	$X(j\omega) = 2\pi\sum_{k=-\infty}^{\infty} X[k]\delta(\omega - k\omega_o)$
$x(t) = \cos(\omega_o t)$	$X(j\omega) = \pi\delta(\omega-\omega_o) + \pi\delta(\omega+\omega_o)$
$x(t) = \sin(\omega_o t)$	$X(j\omega) = \frac{\pi}{j}\delta(\omega-\omega_o) - \frac{\pi}{j}\delta(\omega+\omega_o)$
$x(t) = e^{j\omega_o t}$	$X(j\omega) = 2\pi\delta(\omega-\omega_o)$
$x(t) = \sum_{n=-\infty}^{\infty}\delta(t-nT_s)$	$X(j\omega) = \frac{2\pi}{T_s}\sum_{k=-\infty}^{\infty}\delta\left(\omega - k\frac{2\pi}{T_s}\right)$
$x(t) = \begin{cases} 1, & \lvert t\rvert \le T_o \\ 0, & T_o < \lvert t\rvert < T/2 \end{cases}$ $x(t+T) = x(t)$	$X(j\omega) = \sum_{k=-\infty}^{\infty}\frac{2\sin(k\omega_o T_o)}{k}\delta(\omega - k\omega_o)$

Figure 4: Properties of FT

Property	*Fourier Transform* $x(t) \xleftrightarrow{FT} X(j\omega)$ $y(t) \xleftrightarrow{FT} Y(j\omega)$
Linearity	$ax(t) + by(t) \xleftrightarrow{FT} aX(j\omega) + bY(j\omega)$
Time shift	$x(t-t_o) \xleftrightarrow{FT} e^{-j\omega t_o}X(j\omega)$
Frequency shift	$e^{j\gamma t}x(t) \xleftrightarrow{FT} X(j(\omega-\gamma))$
Scaling	$x(at) \xleftrightarrow{FT} \frac{1}{\lvert a\rvert}X\left(\frac{j\omega}{a}\right)$
Differentiation in time	$\frac{d}{dt}x(t) \xleftrightarrow{FT} j\omega X(j\omega)$
Differentiation in frequency	$-jtx(t) \xleftrightarrow{FT} \frac{d}{d\omega}X(j\omega)$
Integration/ Summation	$\int_{-\infty}^{t} x(\tau)\,d\tau \xleftrightarrow{FT} \frac{X(j\omega)}{j\omega} + \pi X(j0)\delta(\omega)$
Convolution	$\int_{-\infty}^{\infty} x(\tau)y(t-\tau)\,d\tau \xleftrightarrow{FT} X(j\omega)Y(j\omega)$
Multiplication	$x(t)y(t) \xleftrightarrow{FT} \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\nu)Y(j(\omega-\nu))\,d\nu$
Parseval's Theorem	$\int_{-\infty}^{\infty}\lvert x(t)\rvert^2\,dt = \frac{1}{2\pi}\int_{-\infty}^{\infty}\lvert X(j\omega)\rvert^2\,d\omega$
Duality	$X(jt) \xleftrightarrow{FT} 2\pi x(-\omega)$
Symmetry	$x(t)$ real $\xleftrightarrow{FT} X^*(j\omega) = X(-j\omega)$ $x(t)$ imaginary $\xleftrightarrow{FT} X^*(j\omega) = -X(-j\omega)$ $x(t)$ real and even $\xleftrightarrow{FT} \mathrm{Im}\{X(j\omega)\} = 0$ $x(t)$ real and odd $\xleftrightarrow{FT} \mathrm{Re}\{X(j\omega)\} = 0$

Figure 5: Properties of FS

Fourier Series $x(t) \xleftrightarrow{FS;\omega_o} X[k]$ $y(t) \xleftrightarrow{FS;\omega_o} Y[k]$ *Period* $= T$	—
$ax(t) + by(t) \xleftrightarrow{FS;\omega_o} aX[k] + bY[k]$	—
$x(t-t_o) \xleftrightarrow{FS;\omega_o} e^{-jk\omega_o t_o}X[k]$	$\int_0^T x(\tau)y(t-\tau)\,d\tau \xleftrightarrow{FS;\omega_o} TX[k]Y[k]$
$e^{jk_o\omega_o t}x(t) \xleftrightarrow{FS;\omega_o} X[k-k_o]$	$x(t)y(t) \xleftrightarrow{FS;\omega_o} \sum_{l=-\infty}^{\infty} X[l]Y[k-l]$
$x(at) \xleftrightarrow{FS;a\omega_o} X[k]$	$\frac{1}{T}\int_0^T \lvert x(t)\rvert^2\,dt = \sum_{k=-\infty}^{\infty}\lvert X[k]\rvert^2$
$\frac{d}{dt}x(t) \xleftrightarrow{FS;\omega_o} jk\omega_o X[k]$	$x[n] \xleftrightarrow{DTFT} X(e^{j\Omega})$ $X(e^{jt}) \xleftrightarrow{FS;1} x[-k]$

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